

Evaluating Saliency Methods in Medical Vision-Language Models

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Abstract—Visual saliency methods have been developed to explain model decision making, and in a field such as medicine, where decisions can have serious consequences, explainability is crucial for trust and adoption. While saliency methods have been benchmarked on Vision Transformer (ViT) classifiers in medical tasks, they have not been systematically evaluated on medical vision-language models (VLMs). This thesis adapts three explainability methods, Attention Rollout, GMAR, and Grad-CAM, to VLM architectures and evaluates them using a teacher-forced deletion experiment on the ROCov2 radiology dataset with two medical VLMs: MedGemma and LLaVA-Med. The same framework is then used to quantify the extent to which generated medical captions are visually grounded. The deletion results show that, for both models, the saliency methods produce larger probability drops than random masking, suggesting that the methods effectively identify image regions relevant to generated words. Attention Rollout and GMAR perform almost identically across both models, while Grad-CAM generally underperforms the rollout methods. An ablation on MS-COCO, a general captioning dataset, and on the corresponding base models (Gemma 3 and LLaVA-1.5) shows that AOPC is strongly affected by dataset domain, with most models producing larger probability drops on the general captioning dataset. Code is available at: <https://github.com/Belovskiy/Thesis>.

Index Terms—Attention Rollout, Explainable AI (XAI), Vision Transformer, Vision-Language Models, Grad-CAM

I. INTRODUCTION

A key challenge in the widespread clinical adoption of AI models is that they are largely black-box. It is difficult to trust a model with a diagnosis if there is no way to understand why it produced a given output. The field of explainable AI (XAI) has tackled model explainability with saliency methods which are post-hoc algorithms applied to image-processing model architectures to produce heatmaps over the input. These heatmaps are designed to highlight the regions of the image that the model relied on when assigning a particular class, and Figure 1 illustrates an example from the experiments in this thesis. Saliency methods have been developed for many model architectures, with the most recent literature targeting Vision Transformers [1] (ViTs). A ViT adapts the transformer architecture to images by splitting the input into a grid of patches, embedding each patch as a token, and processing the resulting sequence through self-attention layers, with a

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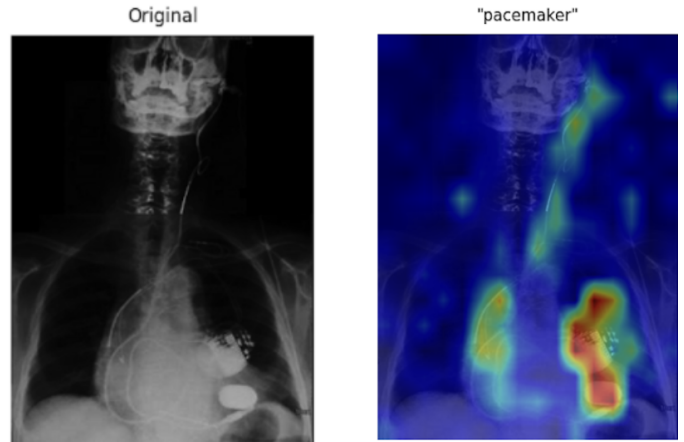


Fig. 1. Example saliency map produced for a single generated token by MedGemma. *Left*: input image. *Right*: Attention Rollout heatmap for the generated token "pacemaker", overlaid on the input. The heatmap localizes onto the pacemaker.

prepended CLS token whose final hidden state is read out by a classification head. The three saliency methods evaluated in this thesis, Gradient-Weighted Class Activation Mapping (Grad-CAM) [2], Attention Rollout [3] and Gradient-driven Multi-head Attention Rollout (GMAR) [4], were all adapted and validated for this setting.

However, when it comes to industry adoption of AI models in medicine, the next frontier is Vision-Language Models (VLMs) rather than classifiers. A VLM is, at the architectural level, a Large Language Model (LLM) coupled with a ViT acting as the vision encoder, enabling multimodal input of image and text into a language model. Several VLMs have been trained for clinical tasks. However, the saliency methods described above were designed and validated for classifiers, and their behavior on medical VLMs has not been systematically evaluated. This study aims to bridge this gap by evaluating how Attention Rollout, GMAR and Grad-CAM perform when applied to two medical VLMs, MedGemma [5] and LLaVA-Med [8], on the ROCov2 [7] radiology dataset. A deletion test is used to evaluate how effectively these methods identify the important regions in the image, and an ablation test was carried out to study the effect of modality on visual

grounding. This thesis aims to answer the following questions:

- 1) How effective are different explainability methods such as GMAR, Grad-CAM, and Attention Rollout at identifying important image regions in medical VLMs?
- 2) To what extent do medical VLMs rely on visual input rather than language priors when generating medical captions?

The novelty of this thesis lies in evaluating token-level saliency faithfulness for autoregressive medical VLMs, rather than for classifier-based vision models. In answering these questions, the contributions made are: (i) an adaptation of Attention Rollout, GMAR, and Grad-CAM from ViT classifiers to specific VLM architectures, (ii) a benchmark of these methods on medical VLMs using a deletion test, (iii) an ablation study to quantify the extent to which generated captions are grounded in the image rather than in language priors. Finally, through this exploration it will become clear if saliency methods can be used to benchmark faithfulness in medical VLMs.

II. RELATED WORK

Saliency methods have been benchmarked on classifiers using both general and, most importantly for this study, medical datasets. Saporta *et al.* [18] benchmark seven saliency methods on chest X-ray pathology classifiers and find that the best methods substantially underperform expert localization. Arun *et al.* [19] report similar findings on classifiers spanning CT, MRI and X-ray. DeGrave *et al.* [20] use saliency methods to discover that COVID-19 radiograph classifiers were attending to image metadata rather than pulmonary signal. Closest to the methods used in this thesis, Barekatin and Glocker [21] quantitatively compare Gradient Attention Rollout and Grad-CAM on several ViT backbones (ViT, DeiT, DINO, Swin) on blood cell and breast ultrasound classification, reporting that Grad-CAM produces more localized and class-discriminative attribution than Gradient Attention Rollout on those tasks.

Explainability for VLMs has largely targeted contrastive or encoder architectures. Chefer *et al.* [23] introduce a relevance-propagation method for bi-modal and encoder-decoder transformers, and MM-SHAP [24] measures the relative contribution of each modality to a VLM prediction. Neither is designed for per-token attribution in an autoregressive text generator, which is the setting studied in this thesis.

Two gaps therefore emerge. First, the medical saliency studies above all target *classifiers* that emit a single class logit, whereas clinical adoption is moving toward generative VLMs that produce free-text documents; the attribution target and the architecture the signal must propagate through are different. Second, VLM explainability has rarely been evaluated *per generated token* on autoregressive medical caption generators. This thesis addresses both gaps: it adapts Attention Rollout, GMAR and Grad-CAM to the causal-decoder setting, evaluates them per generated token with a teacher-forced deletion test, and uses the same framework to study how visually grounded the generated captions are. The two research questions are thus the generative-VLM counterparts of the faithfulness benchmarks established for classifiers [18],

[21] and the grounding concerns recently raised for medical VLMs [22].

III. METHODS

A. Adapting Saliency Methods to VLMs

At the architectural level, the VLMs studied here share a common pattern: a ViT vision encoder produces per-patch embeddings that a projector maps into the hidden space of an autoregressive LLM, where they are inserted as tokens alongside the text tokens. The three saliency methods were originally formulated for ViT classifiers, where they target a single class logit produced by the classifier. When porting these methods to generative VLMs, three adaptations are required:

- 1) The target is a per-token next-token logit at a specific position p in the generated sequence, rather than a per-class output logit. A separate saliency map is produced for each generated token.
- 2) The attention matrices and intermediate activations used by the methods come from the LLM’s causal decoder rather than from a bidirectional ViT encoder.
- 3) The saliency for the token at position p is read as the row of the resulting relevance matrix at position p , restricted to the sequence positions occupied by image-patch tokens, and reshaped to the spatial grid implied by the patch count (24×24 for LLaVA-Med, 16×16 for MedGemma).

a) Method selection: The three methods were chosen to span the main families of post-hoc attribution in transformer architectures: pure attention (Attention Rollout), pure gradient (Grad-CAM), and a gradient-weighted hybrid (GMAR). They are also the methods most recently adapted and validated for ViTs [4], [21], which makes them the natural candidates for testing whether state-of-the-art ViT saliency methods transfer to the VLM setting.

b) Attention Rollout and GMAR: Both methods compose per-layer attention matrices across a uniformly spaced subset of decoder layers $\mathcal{L}$ (layers $\{7, 15, 23, 31\}$ for LLaVA-Med and $\{5, 11, 17, 23, 29\}$ for MedGemma; see Appendix A) to mitigate attention collapse. For layer l with H total attention heads, the combined attention matrix $A^{(l)}$ is a weighted sum:

$$A^{(l)} = \sum_{h=1}^H w_h^{(l)} A_h^{(l)}$$

Here, $A_h^{(l)}$ represents the attention matrix for an individual head h , and $w_h^{(l)}$ denotes its assigned weight. In Attention Rollout, heads are weighted uniformly ($w_h^{(l)} = 1/H$). GMAR derives $w_h^{(l)}$ by computing the gradient of the next-token logit $y_t^{(p)}$ (the prediction for target ID t at position p) with respect to each head ($G_h^{(l)}$), calculating its L1 or L2 norm, retaining only non-negative values, and L1-normalizing across heads. The rollout step comes from blending $A^{(l)}$ with the identity matrix and row-normalizing. The cumulative rollout matrix A_{rollout} is the matrix product of these adjusted layers over $\mathcal{L}$.

Saliency for token p is extracted from $A_{\text{rollout}}[p, \cdot]$ at the image-patch columns.

c) **Grad-CAM**: Grad-CAM targets the activation matrix $A \in \mathbb{R}^{N \times D}$ hooked at the multimodal projector output. For the generated token at position p with target ID t , importance for each patch position $i \in \{1, \dots, N\}$ is computed as:

$$L_i = \text{ReLU} \left(\sum_{k=1}^D \frac{\partial y_t^{(p)}}{\partial A_{i,k}} \cdot A_{i,k} \right)$$

Here, N defines the total number of visual patch embeddings, D represents the embedding dimension size, and $A_{i,k}$ is the specific activation value for patch i at dimension k with ReLU isolating positive contributions to the prediction.

B. Evaluation Metric: Perturbation via Teacher Forcing

To quantitatively evaluate the faithfulness of the generated heatmaps, this thesis uses the Area Over the Perturbation Curve (AOPC) [16]. AOPC measures the decline in a model’s prediction confidence as the most salient visual features, identified by the respective explainability method, are progressively removed.

AOPC is chosen because it is a deletion-based faithfulness metric: it directly tests whether the regions a method identifies as important are causally responsible for the prediction, by measuring how much the prediction degrades when those regions are removed. Also, AOPC requires no ground-truth region annotations, and it reduces a full perturbation curve to a single value that is comparable across methods and models. Additionally, it is a widely used metric in the XAI literature.

As VLMs are autoregressive, the probability of predicting a token depends on its position in the sequence. Unbounded generation allows the model to continuously alter its output sequence, making token-level attribution unstable. To isolate the model’s internal state, teacher forcing [17] is applied. Given an original image I and a generated sequence of tokens $S = \{t_1, t_2, \dots, t_n\}$, the preceding tokens $S_{<i} = \{t_1, t_2, \dots, t_{i-1}\}$ are provided to the model as a fixed language prior.

Perturbations are applied via a zero¹ masking strategy. For a given token t_i and a masking ratio r , the top r fraction of the most salient regions is replaced to create a perturbed image, denoted as $\tilde{I}_{i,r}$. The probability drop, $\Delta P(t_i, r)$, is then calculated as the difference between the initial probability of generating token t_i and the probability of generating t_i given the perturbed image:

$$\Delta P(t_i, r) = P(t_i | I, S_{<i}) - P(t_i | \tilde{I}_{i,r}, S_{<i})$$

The final AOPC is calculated as the average probability drop across a predefined sequence of masking ratios.

IV. EXPERIMENTS

Two experiments were conducted: the main experiment evaluating the explainability methods for the two medical VLMs, and an ablation experiment which involved the two

¹Zero masking means replacing the selected image patches with zero-valued pixels before re-running the model.

medical VLMs as well as their base models, LLaVA-1.5 [13] and Gemma 3 [6]. In the ablation experiment, two datasets were used: ROCoV2 and MS-COCO [12], a general captioning dataset. Both experiments were conducted on an NVIDIA RTX PRO 6000 GPU with 96 GB of VRAM. The sample included 500 images from the ROCoV2 dataset and 100 from MS-COCO.

A. Models

MedGemma is a clinical VLM developed by Google, built on the Gemma 3 multimodal language model backbone; its language model is a 34-layer decoder and its vision encoder is MedSigLIP, a SigLIP-2 [10] variant pre-trained on medical images. LLaVA-Med is an open-source VLM derived from LLaVA-1.5 and adapted for biomedical visual question answering; the version used here (LLaVA-Med v1.5, Mistral-7B variant) couples a CLIP ViT-L/14 [11] visual encoder with a 32-layer Mistral-7B [9] autoregressive language model.

B. Evaluating Explainability Methods

The medical VLMs received an image I from the ROCoV2 dataset alongside the prompt: “Generate a single-sentence medical caption for this radiology image”. To evaluate the faithfulness of the generated heatmaps for every token t_i across the output sequence, the AOPC metric described in Section III-B was applied. Perturbations were applied using masking ratios $r \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$. In addition, a random map was generated and perturbed in the same manner to produce a random baseline for comparison.

C. Ablation Experiment

To disentangle the contribution of the medical fine-tuning and medical language to the probability drops, the same perturbation procedure as in Section IV-B was run on a 2×2 ablation: each medical VLM and its non-medical base (MedGemma vs. Gemma 3; LLaVA-Med vs. LLaVA-1.5) were evaluated on both ROCoV2 and MS-COCO. In the MS-COCO condition, the prompt was: “Generate a caption for this image.”

D. Data Processing

a) **Word Reconstruction From Tokens**: Both MedGemma and LLaVA-Med employ subword tokenization, which splits words into multiple tokens, for example, “tomography” can be tokenized as `tom`, `ography`. In a subword sequence like `tom` and `ography`, the true probability drop occurs at `tom`, since `ography` is predicted with a probability of 1. To obtain more truthful saliency scores, these sub-tokens are merged into whole words whose boundaries are detected via the tokenizer’s word-initial markers. Punctuation tokens are removed before statistical analysis, as punctuation is a language artifact.

b) **Clinical Named-Entity Recognition**: To assign each reconstructed word a semantically meaningful label, a named entity recognition (NER) model is applied. This thesis uses Clinical-AI-Apollo/Medical-NER [14], a DeBERTa-v3 [15]-based token classifier fine-tuned on biomedical and clinical corpora to predict medical entity types. Each merged word is

then aligned to an entity group. Words that do not overlap any entity (function words, stopwords, residual tokens) are tagged as `OTHER`. This produces a single categorical label per word, which is subsequently used to stratify the faithfulness analysis by semantic class and to see whether saliency methods preferentially attribute probability mass to clinically relevant words. **Words tagged as `OTHER` by the NER are removed before data analysis.**

V. RESULTS

A. Method Faithfulness

Table I reports the mean per-word probability drop. For MedGemma the four informed methods clear the random baseline by roughly a factor of two on AOPC, and by almost six times at the lowest mask ratio. The three rollout methods are clustered together and all exceed Grad-CAM. Figure 2 shows the mean drop rising with the mask ratio r , with the Grad-CAM-rollout gap smallest at $r \in [0.1, 0.2]$ and widening as r grows. For LLaVA-Med the drops are far smaller and the informed methods sit close to random; as r grows they converge towards the baseline, most strongly on LLaVA-Med, where at $r = 0.5$ the rollout family (0.075) is within 7% of random (0.070), whereas on MedGemma the gap is still 0.156 vs. 0.103 (51%). Figure 5 in the appendix shows example maps from each method on a single sample. Figure 2 contains the lines for all saliency methods; however, they are difficult to make out due to the tight clustering among the rollout methods.

TABLE I
MEAN PER-WORD PROBABILITY DROP $\overline{\Delta p}$ BY XAI METHOD AND MASK RATIO FOR MEDGEMMA (TOP) AND LLaVA-MED (BOTTOM).

MedGemma						
Method	0.1	0.2	0.3	0.4	0.5	AOPC
Attention	0.0941	0.1142	0.1280	0.1414	0.1562	0.1268
GMAR-L1	0.0938	0.1143	0.1281	0.1418	0.1567	0.1269
GMAR-L2	0.0936	0.1136	0.1275	0.1409	0.1560	0.1263
Grad-CAM	0.0865	0.1068	0.1190	0.1294	0.1401	0.1164
Random	0.0161	0.0348	0.0572	0.0788	0.1031	0.0580
LLaVA-Med						
Method	0.1	0.2	0.3	0.4	0.5	AOPC
Attention	0.0205	0.0362	0.0508	0.0648	0.0750	0.0494
GMAR-L1	0.0201	0.0358	0.0505	0.0648	0.0753	0.0493
GMAR-L2	0.0203	0.0359	0.0507	0.0648	0.0755	0.0494
Grad-CAM	0.0195	0.0337	0.0419	0.0493	0.0579	0.0404
Random	0.0121	0.0253	0.0397	0.0530	0.0703	0.0401

B. Per-Word Drops Are Heavily Right-Skewed

The mean drops hide a strongly skewed per-word distribution (Figure 4). At $r = 0.1$ the median MedGemma Attention Rollout & GMAR word drops by only ≈ 0.02 , while the mean is ≈ 0.09 and the 95th percentile is ≈ 0.55 ; the most salient tenth of words accounts for about half of the total probability drop (Figure 3). On LLaVA-Med the skew is more extreme (median ≈ 0.003 , mean ≈ 0.02), with the top tenth of

words contributing roughly two thirds of the total. The rollout performance gain over Grad-CAM is confined to the upper tail but at the median the two are indistinguishable, and the rollout variants remain nearly identical across the whole distribution (Figure 4).

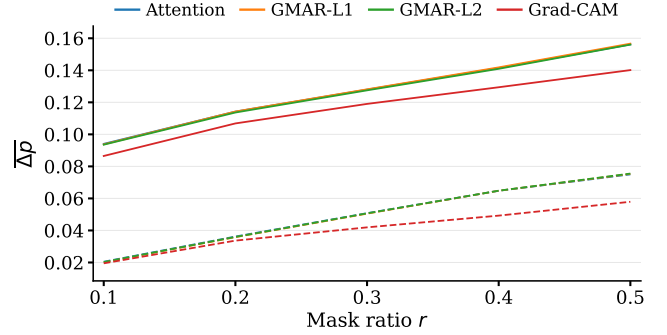


Fig. 2. Mean per-word probability drop $\overline{\Delta p}$ versus mask ratio r for every informed method on both models. Solid lines: MedGemma. Dashed lines: LLaVA-Med.

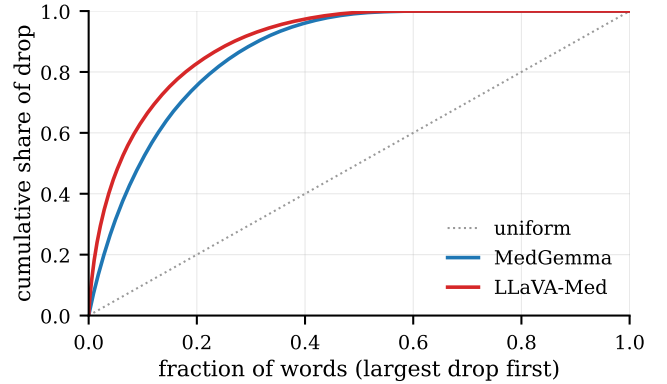


Fig. 3. Cumulative share of the total per-word probability drop versus the fraction of words, with words sorted by their drop (largest first); Attention Rollout, $r = 0.1$. The diagonal marks perfectly uniform grounding; the 10% of words with the largest drops carry 51% (MedGemma) and 64% (LLaVA-Med) of the total.

C. Grounding Concentrates in Specific Clinical Words

We can observe that the tail is semantically structured. Stratifying by NER class (Table II), the large MedGemma drops concentrate in a few classes: `SEVERITY` (≈ 0.28), `THERAPEUTIC_PROCEDURE` (≈ 0.23) and `DETAILED_DESCRIPTION` (≈ 0.21). Most other classes sit near the low-drop bulk. Attention Rollout, GMAR-L1 and GMAR-L2 are near-identical within every class, and Grad-CAM falls behind precisely on the high-drop classes (e.g. `THERAPEUTIC_PROCEDURE`), consistent with its weaker upper tail (Figure 4). On LLaVA-Med the same classes are far smaller: `BIOLOGICAL_STRUCTURE` is the largest for the rollout family (≈ 0.071), `SIGN_SYMPTOM` is near zero and `SEVERITY` is slightly negative. The strongly grounded words are thus the clinically contentful ones, while

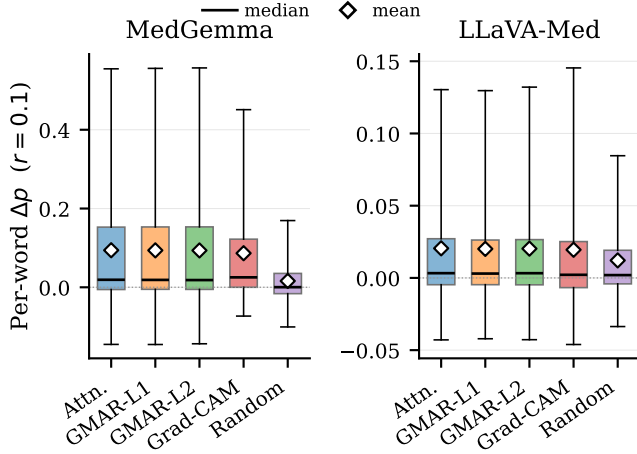


Fig. 4. Distribution of per-word probability drops Δp at $r = 0.1$ by method, for MedGemma (left) and LLaVA-Med (right). Boxes show the median and interquartile range, whiskers the 5th–95th percentile, and diamonds the mean. The mean lies far above the median and the upper whiskers are long, reflecting a strongly right-skewed, tail-dominated distribution.

methodology and templatic words form the near-zero bulk that dominates the count.

TABLE II
AOPC BY NER ENTITY CLASS FOR EACH INFORMED METHOD.

MedGemma				
Entity class	Attention	GMAR-L1	GMAR-L2	Grad-CAM
Bio. Struct.	0.131	0.131	0.131	0.127
Diag. Proc.	0.042	0.043	0.042	0.043
Therap. Proc.	0.233	0.232	0.231	0.101
Det. Desc.	0.207	0.207	0.205	0.170
Dis./Disorder	0.086	0.086	0.085	0.081
Sign/Sympt.	0.104	0.104	0.105	0.099
Severity	0.279	0.278	0.278	0.235
Lab Value	0.100	0.097	0.097	0.091

LLaVA-Med				
Entity class	Attention	GMAR-L1	GMAR-L2	Grad-CAM
Bio. Struct.	0.071	0.071	0.071	0.059
Diag. Proc.	0.034	0.034	0.034	0.027
Therap. Proc.	0.065	0.065	0.064	0.044
Det. Desc.	0.045	0.046	0.046	0.036
Dis./Disorder	0.032	0.032	0.032	0.030
Sign/Sympt.	0.009	0.009	0.009	0.010
Severity	-0.003	-0.003	-0.003	-0.003
Lab Value	0.020	0.020	0.021	0.028

D. Statistical Significance

To test whether the five methods produce different per-word drops, a Friedman test (a non-parametric, paired, repeated-measures test that makes no normality assumption) is applied within each (mask ratio, NER class) stratum, ranking the methods per word and asking whether the rank sums differ more than expected by chance. This yields 50 strata on MedGemma (10 classes $\times$ 5 ratios) and 55 on LLaVA-Med (11

classes); after Benjamini–Hochberg correction at $q < 0.05$ the test rejects equality of methods in 41/50 strata on MedGemma and 27/55 on LLaVA-Med.

E. Base-Model Ablation

Table III reports the Attention Rollout AOPC across the 2×2 ablation. The Gemma-family models and LLaVA-1.5 show larger drops on MS-COCO than on ROCov2, while LLaVA-Med is an exception, with a very small COCO AOPC that, as discussed in Section VI, partly reflects poor generation quality under the MS-COCO prompt.

TABLE III
ATTENTION ROLLOUT AOPC FOR MEDICAL VLMS AND THEIR BASE MODELS ON ROCov2 AND COCO, COMPUTED ON THE FIRST 100 SAMPLES FOR EACH DATASET. ALL VALUES USE WORD-LEVEL DROPS; FOR THE COCO CONDITION, WHERE MEDICAL NER DOES NOT APPLY, PURE PUNCTUATION AND STOPWORDS ARE REMOVED INSTEAD.

Dataset	Attention Rollout AOPC			
	MedGemma	Gemma 3	LLaVA-Med	LLaVA-1.5
ROCov2	0.1105	0.1757	0.0515	0.0674
COCO	0.2802	0.3058	0.0120	0.1490

VI. DISCUSSION

A. Evaluating Explainability Methods

Under AOPC, Attention Rollout, GMAR, and Grad-CAM all identify image regions that are more relevant than random masking (Table I), so the first research question is answered in the affirmative: the methods recover image evidence for the generated tokens, clearly on MedGemma and weakly on LLaVA-Med. However, AOPC has two main problems. Firstly, AOPC is less discriminative at high mask ratios, where removing a large fraction of the image makes different heatmaps overlap, so the lower ratios are more informative, and there the informed methods exceed random by roughly two times on LLaVA-Med and almost six times on MedGemma. Secondly, AOPC is a mean, and the per-word distribution shows that this mean is carried by a minority of words: the median drop is near zero, while a long upper tail of strongly grounded words pulls the mean up (Figure 4). Attention Rollout and GMAR are indistinguishable, while Grad-CAM matches them on the near-zero median but falls behind on the upper tail (Figure 4) and, equivalently, on the high-drop clinical classes (Table II), so its deficit is specific to the strongly grounded words that matter clinically, which is an informative result.

The AOPC performance ordering differs from deletion results obtained on ViTs, where GMAR can outperform both Attention Rollout and Grad-CAM, and Grad-CAM can outperform Attention Rollout [4]. However, the results obtained in this thesis are not directly comparable to prior ViT studies. This is due to the difference in datasets, masking ranges, which in most ViT studies span $r \in [0, 1]$ [4], [21], and the use of teacher-forced word probabilities rather than class logits. Furthermore, the deletion scores between saliency methods in ViT studies are very close (within a few percent), which is the

same behavior seen here. Most importantly, ViTs and VLMs are fundamentally different models so it is to be expected that these saliency methods will behave differently. The difference in behavior stems from the effectiveness of gradients; we can see that a pure attention method (Attention Rollout) performs the same as a gradient-weighted attention method (GMAR) and that a pure gradient method (Grad-CAM) underperforms the two. This suggests that gradients are less effective in VLMs than in ViTs. This is due to gradient shattering which is a phenomenon that occurs in deep-autoregressive transformers. Below is a breakdown of how this phenomenon affects the saliency methods explored in ViTs and VLMs.

a) Attention Rollout and GMAR: Attention Rollout is the unweighted, naive case of GMAR: it averages all attention heads equally while GMAR weighs them by a gradient-based importance score. In ViTs, gradient weighting leads to a performance gain as the attribution path is relatively direct: gradients are taken from the class score back through a bidirectional image encoder, where patch tokens are the main representational unit and different heads capture distinct spatial relations. As a result, the gradient norms assigned to attention heads can remain meaningfully non-uniform. On the other hand, in VLMs, the target is a token probability and the attribution signal must propagate through a deep causal language decoder before reaching the image-token positions. During this propagation, the gradient signal is repeatedly mixed across many preceding text and image positions by causal self-attention with the result being that the per-head gradient norms converge, after normalization to the unweighted $1/H$ scalar. Thus, Attention Rollout and GMAR become nearly identical, which explains their near identical performance in the deletion test.

b) Grad-CAM: The same gradient-flattening shattering also contributes to Grad-CAM’s underperformance. Grad-CAM relies entirely on the gradient of the target logit with respect to a hooked feature map. In this setup, that gradient must traverse the causal decoder, where each layer aggregates information from earlier positions. This reduces the localization of the per-position attribution signal. The effect may be compounded by the hook location: the multimodal projector output is already downstream of CLIP or MedSigLIP image encoding, where image patch information has been mixed by bidirectional attention layers. As a result, gradients at the projector do not necessarily map cleanly back to individual image regions.

Different results could have been obtained by selecting another Grad-CAM hook target or another subset of decoder layers for the rollout methods. The observations above therefore apply to this experimental setup and should not be treated as a general statement regarding all (medical) VLMs of similar architecture.

B. Evaluating Grounding

The second research question is answered as follows: the models exhibit statistically significant but limited visual grounding, and what grounding exists is heavily concentrated

in a small minority of clinically contentful tokens. The bulk of a medical caption’s content is methodology, anatomy, and templatic phrasing, which is directly predictable from a perfect language prior, so the marginal contribution of image content to next-token prediction is inherently smaller for medical captions than for general-domain ones.

This language-prior effect is directly visible in the ablation experiment results (Table III): three of the four models produced AOPC values roughly two to three times higher on MS-COCO than on ROCov2. LLaVA-Med is the exception. An inspection of its generated MS-COCO outputs shows that many of its captions were not clean image descriptions: 23 of the 100 captions are a recital of LLaVA-Med’s system prompt, and some degenerate into repeated fragments. The low LLaVA-Med COCO AOPC should therefore be interpreted as a reflection of generation quality under this prompt, rather than purely as a reflection of visual grounding.

Within the medical captions, the concentration is extreme: approximately one word in ten accounts for more than half of the total probability drop, and those words are the clinically contentful ones (Table II), while methodology and templatic words contribute almost nothing.

C. Societal Implications

In an ideal case, a medical AI model would never report a finding that is not grounded in the visual material it was presented with. However, the language-prior effect observed here undermines that ideal: when the visual signal is weakened, both VLMs fall back on the priors that produced the reference caption. This has a direct consequence for deployment. If a model is used conversationally, and the surrounding context contains information about other patients or images, its output becomes prone to reflecting that context rather than the image at hand, increasing the risk of ungrounded findings in a setting where errors carry serious consequences. This aligns with recent work that emphasizes evaluating medical VLMs on spatial grounding rather than on benchmark accuracy alone [22], which reports that a VLM answered anatomical relationship questions almost as accurately with no image as with the image, which is a direct symptom of the same prior-driven behavior. The findings in this thesis therefore support the need for grounding to be a first-class benchmark for medical VLMs, and suggest that visual saliency methods can be used for that purpose.

D. Limitations

a) Teacher forcing: Using teacher forcing means that this experiment does not measure free-form reliance on the image. Rather, it measures the marginal contribution of image content to next-token prediction, conditional on the model having perfect prior context. Giving a medically fine-tuned model a prior such as “... that shows acute” and expecting anything other than “appendicitis” would not be a meaningful test of visual grounding, regardless of the image. When the visual signal is weakened by perturbation, both VLMs default to the language priors that produced the reference caption, so the AOPC drop

is bounded above by the margin the image contributes over those priors. This bound may be tighter on LLaVA-Med than on MedGemma, because LLaVA-Med’s Mistral-7B backbone was trained as a text-only model [9] whereas MedGemma’s Gemma 3 backbone was trained multimodally from the start; MedGemma is therefore expected to rely more directly on visual tokens, consistent with its substantially larger AOPC values.

b) Inter-model comparison and generalization: A direct comparison between LLaVA-Med and MedGemma is not meaningful: the models differ substantially despite sharing the same overall architecture. MedGemma is newer (2025) than LLaVA-Med (2023) and has a medically fine-tuned ViT encoder, whereas LLaVA-Med was a proof-of-concept showing LLaVA-1.5 could be adapted to the biomedical domain on a small training budget [8]. The two should not be treated as equally representative data points, and no generalization to the whole field of medical VLMs should be made on the basis of these two models.

c) Prompt and model conditioning: Using the same prompt within each dataset condition facilitates standardization, but it also means the measured performance is a property of the model–prompt pair rather than of the model alone. The results should be interpreted as one prompting protocol, not the best grounding each model can achieve. Specialized prompts tailored to a model’s known behavior can lead to higher AOPC scores.

d) Data leakage and dataset artifacts: MedGemma or LLaVA-Med may have seen some ROCov2 images, or visually similar images, during training. A memorized image–caption pair could be reproduced from memory rather than from visual evidence, reducing measured drops. ROCov2 also contains visible annotations, labels, arrows and text markers: an attribution method may correctly flag a visible “liver” label, yet after masking it the model can still infer the word from surrounding anatomy or caption context, hence the drop stays small. The deletion metric can therefore underestimate shortcut-like evidence when redundant cues remain, which relates to prior findings where models exploit shortcuts over clinically meaningful features [20].

E. Future Work

Several directions extend this work. A complementary protocol that generates captions with and without the image, rather than teacher forcing, would measure overall image dependence instead of the marginal contribution measured here. Model-specific conditioning strategies should be compared against a shared-prompt baseline, and the same methods should be evaluated on images with annotation artifacts removed or stratified by the presence of visible annotations. Methodologically, future work should test alternative Grad-CAM hook locations and decoder-layer subsets, directly measure the per-head gradient distributions that lead to Attention Rollout–GMAR equivalence, and complement the deletion test with an insertion test for a more complete picture.

VII. CONCLUSION

This thesis empirically benchmarked saliency methods for medical vision-language models. Attention Rollout, GMAR, and Grad-CAM were adapted from ViT classifiers to VLMs to generate one saliency map per generated token, and were evaluated on MedGemma and LLaVA-Med with a teacher-forced deletion test on ROCov2.

RQ1: All three methods produced larger probability drops than a random baseline, so they do identify image regions that contribute to the generated tokens. The effect is pronounced on MedGemma and weak on LLaVA-Med. Attention Rollout and GMAR were nearly indistinguishable, indicating that gradient-based head weighting is ineffective in this setting, and Grad-CAM underperformed the rollout methods, consistent with weak gradient signals through the causal decoder.

RQ2: The models showed statistically significant but limited visual grounding, bounded by strong language priors. The ablation showed that this grounding depends on model training, prompt, and dataset domain, with larger probability drops on the general-domain dataset.

DECLARATION ON GENERATIVE AI

In adherence to the disclosure guidelines proposed by M. Dumontier [25], I disclose my use of generative AI assistance for the following tasks:

- 1) Debugging, optimizing, and documenting the code used to answer the research questions. Models: Claude Opus 4.6 and Claude Sonnet 4.6.
- 2) Drafting and proofreading parts of the thesis text for grammar and spelling. Models: Kimi 2.6 and Claude Sonnet 4.6.
- 3) Generating and formatting figures and tables to fit the IEEE template. Model: Claude Sonnet 4.6.

REFERENCES

- [1] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, “An image is worth 16x16 words: Transformers for image recognition at scale,” in *Proc. Int. Conf. Learning Representations (ICLR)*, 2021. arXiv:2010.11929.
- [2] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, “Grad-CAM: Visual explanations from deep networks via gradient-based localization,” in *Proc. IEEE Int. Conf. Computer Vision (ICCV)*, 2017, pp. 618–626. arXiv:1610.02391.
- [3] S. Abnar and W. Zuidema, “Quantifying attention flow in transformers,” in *Proc. 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2020, pp. 4190–4197. arXiv:2005.00928.
- [4] S. Jo, G. Jang, and H. Park, “GMAR: Gradient-driven multi-head attention rollout for vision transformer interpretability,” arXiv preprint arXiv:2504.19414, 2025.
- [5] A. Sellergren *et al.*, “MedGemma technical report,” arXiv preprint arXiv:2507.05201, 2025.
- [6] Gemma Team, Google DeepMind, “Gemma 3 technical report,” arXiv preprint arXiv:2503.19786, 2025.
- [7] J. Rückert *et al.*, “ROCov2: Radiology objects in COntext version 2, an updated multimodal image dataset,” *Scientific Data*, vol. 11, no. 1, p. 688, 2024. arXiv:2405.10004.
- [8] C. Li, C. Wong, S. Zhang, N. Usuyama, H. Liu, J. Yang, T. Naumann, H. Poon, and J. Gao, “LLaVA-Med: Training a large language-and-vision assistant for biomedicine in one day,” in *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2023. arXiv:2306.00890.

- [9] A. Q. Jiang *et al.*, “Mistral 7B,” arXiv preprint arXiv:2310.06825, 2023.
- [10] M. Tschannen *et al.*, “SigLIP 2: Multilingual vision-language encoders with improved semantic understanding, localization, and dense features,” arXiv preprint arXiv:2502.14786, 2025.
- [11] A. Radford *et al.*, “Learning transferable visual models from natural language supervision,” in *Proc. Int. Conf. Machine Learning (ICML)*, 2021, pp. 8748–8763. arXiv:2103.00020.
- [12] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, and C. L. Zitnick, “Microsoft COCO: Common objects in context,” in *Proc. European Conf. Computer Vision (ECCV)*, 2014, pp. 740–755. arXiv:1405.0312.
- [13] H. Liu, C. Li, Y. Li, and Y. J. Lee, “Improved baselines with visual instruction tuning,” in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2024. arXiv:2310.03744.
- [14] Clinical-AI-Apollo, “Medical-NER: A DeBERTa-v3 token classifier fine-tuned on PubMed for 41 medical entity types,” Hugging Face model repository, 2024. [Online]. Available: <https://huggingface.co/Clinical-AI-Apollo/Medical-NER>.
- [15] P. He, J. Gao, and W. Chen, “DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing,” in *Proc. Int. Conf. Learning Representations (ICLR)*, 2023. arXiv:2111.09543.
- [16] W. Samek, A. Binder, G. Montavon, S. Lapuschkin, and K.-R. Müller, “Evaluating the visualization of what a deep neural network has learned,” *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 28, no. 11, pp. 2660–2673, 2017.
- [17] R. J. Williams and D. Zipser, “A learning algorithm for continually running fully recurrent neural networks,” *Neural Computation*, vol. 1, no. 2, pp. 270–280, 1989.
- [18] A. Saporta, X. Gui, A. Agrawal, A. Pareek, S. Q. H. Truong, C. D. T. Nguyen, V.-D. Ngo, J. Seekins, F. G. Blankenberg, A. Y. Ng, M. P. Lungren, and P. Rajpurkar, “Benchmarking saliency methods for chest X-ray interpretation,” *Nature Machine Intelligence*, vol. 4, no. 10, pp. 867–878, 2022.
- [19] N. Arun, N. Gaw, P. Singh, K. Chang, M. Aggarwal, B. Chen, K. Hoebel, S. Patel, M. Gidwani, J. Adebayo, M. D. Li, and J. Kalpathy-Cramer, “Assessing the trustworthiness of saliency maps for localizing abnormalities in medical imaging,” *Radiology: Artificial Intelligence*, vol. 3, no. 6, p. e200267, 2021.
- [20] A. J. DeGrave, J. D. Janizek, and S.-I. Lee, “AI for radiographic COVID-19 detection selects shortcuts over signal,” *Nature Machine Intelligence*, vol. 3, no. 7, pp. 610–619, 2021.
- [21] L. Barekatain and B. Glocker, “Evaluating the Explainability of Vision Transformers in Medical Imaging,” in *Workshop on Interpretability of Machine Intelligence in Medical Image Computing (iMIMIC), MICCAI 2025*, 2025. arXiv:2510.12021.
- [22] A. S. Yu, M. Hariri, K. Nakamura, M. Yang, X. Li, and V. Chaudhary, “Medical image spatial grounding with semantic sampling,” arXiv preprint arXiv:2603.14579, 2026.
- [23] H. Chefer, S. Gur, and L. Wolf, “Generic attention-model explainability for interpreting bi-modal and encoder-decoder transformers,” in *Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV)*, 2021, pp. 397–406. arXiv:2103.15679.
- [24] L. Parcalabescu and A. Frank, “MM-SHAP: A performance-agnostic metric for measuring multimodal contributions in vision and language models & tasks,” in *Proc. 61st Annual Meeting of the Association for Computational Linguistics (ACL)*, 2023, pp. 4032–4059. arXiv:2212.08158.
- [25] M. Dumontier, “Policy on the Use of Artificial Intelligence in Education,” Mar. 17, 2026. doi: 10.5281/zenodo.19074298.

APPENDIX

Exact model checkpoints used in the main and ablation experiments.

TABLE IV
MODEL CHECKPOINTS (HUGGING FACE) USED IN THIS STUDY.

Model	Hugging Face checkpoint
MedGemma 4B	google/medgemma-1.5-4b-it
LLaVA-Med 7B	microsoft/llava-med-v1.5-mistral-7b
Gemma 3 4B (base)	google/gemma-3-4b-it
LLaVA-1.5 7B (base)	llava-hf/llava-1.5-7b-hf

Code availability. The code implementation is available at <https://github.com/Bel0vskiy/Thesis>.

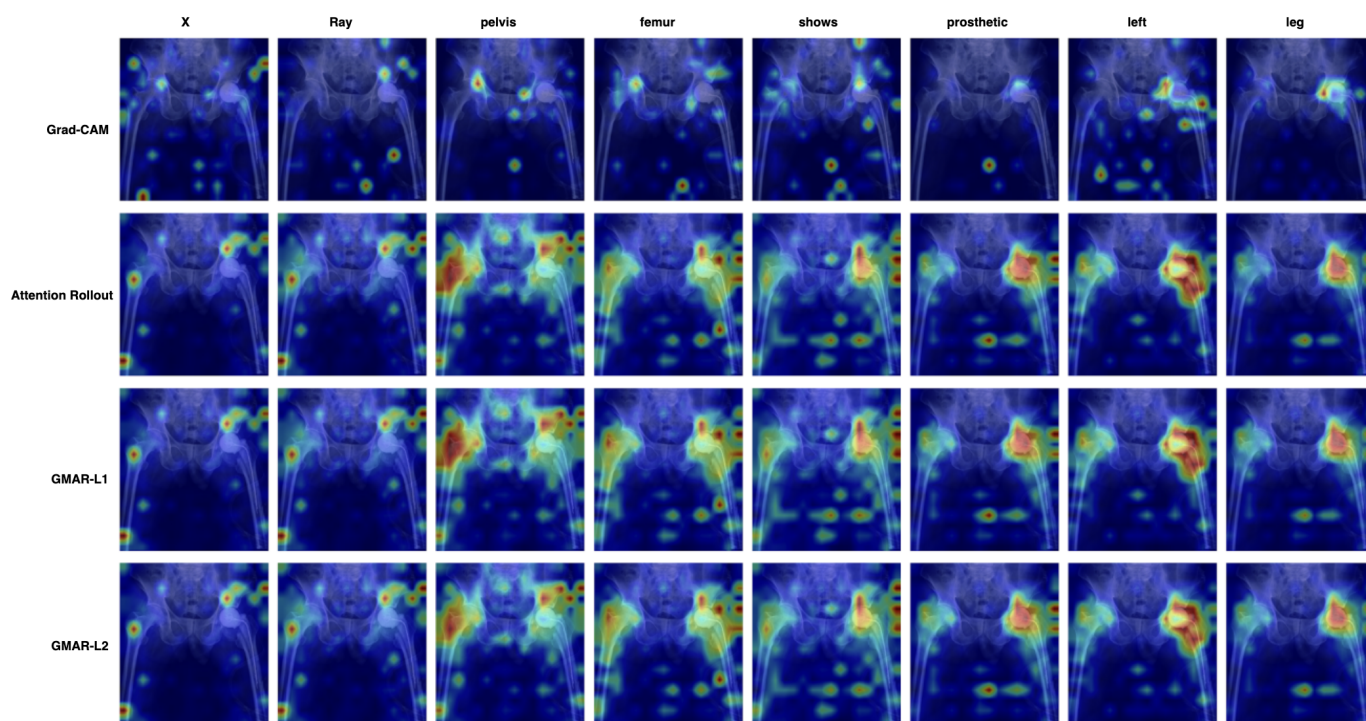


Fig. 5. Example saliency maps generated by each method for a random ROCOv2 sample with stopwords removed.